



Addressing Sustainability Obstacles, Financial Trade-offs and Key Performance Indicators Across Wind Turbine Life Cycle

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Abstract

The goal of this thesis was to examine sustainability challenges and the trade-offs in life cycle of wind turbines are analyzed in this thesis, taking into account both the financial and sustainability Key Performance Indicators (KPIs). Key environmental and economic impacts are identified and discussed across the life cycle, which is split in 4 main stages: raw material extraction, manufacturing and installation, operation and maintenance, and decommissioning and recycling. The findings emphasize that raw material extraction and manufacturing present the most emission-intensive stages, due to vast use of steel, concrete, and composite material. On the other hand, the operational phase is less emission-intensive, but presents logistical challenges, mainly in the case of offshore turbines. Lastly, decommissioning faces significant limitations with regard to the recycling of composite rotor blades that often end up in landfills. Sustainability KPIs are used for assessing environmental impacts (e.g. Carbon Footprint (CF) or Energy Return on Investment (EROI)), while financial KPIs are used for assessing economic viability (e.g. Levelized Cost of Energy (LCOE) and Operational Expenditures (OPEX)). There are obvious trade-offs between cost-effectiveness and environmental goals, especially in the case of material procurement and recycling. Interviews with industry experts revealed possible opportunities for innovation, such as low-emission steel and technologies for recycling. These interviews have shown that the findings in the literature are not totally aligned with the current state in the industry.

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Introduction

1.1 Problem Statement

1.1.1 Brief Historical Overview

While mankind has been harnessing the wind power for over 3 millennia, it has only started using wind potential to generate electrical energy within last century and a half [1]. Towards the end of 19th century Charles F. Brush (1849–1929) laid out plans and constructed in Cleveland, USA a first wind turbine, which at that time was 17m in diameter and consisted of 144 blades [2].

In the following 20th century, advancements in wind energy field can be split up into two distinct intervals. Between the beginning of 1900 and 1973, wind-generated electricity was not viable, in terms of the production costs. Main reasons for avoiding the wind turbines for production of energy included expansion of electrical grids and the usage of inexpensive fossil fuels [1].

However, turning point for the wind industry were the Oil Crises of 1973 and 1979, marked by disruptions in the supply-chain and volatile fossil fuel prices, which accelerated the shift toward alternative energy sources. Consequently, European countries and the USA emerged as pioneers in the field of wind energy, as they decided to invest and subsidy the research on extensive commercial wind turbines. Following this trend, in 1978, the world's first multi-megawatt wind turbine made its debut, setting a standard which many are still being followed by modern wind turbines. Interestingly, towards the end of 1980s, NASA made a breakthrough by constructing wind turbines with capacities of 3.2 MW and 4 MW. Nevertheless, as oil prices declined again, wind-generated electricity became less economically competitive, leading many manufacturers to again exit the market and opt for others [1].

New decisive moment came in the early 21st century, as the topics such as growing concerns over energy security, global warming, and eventual fossil fuel depletion renewed

the focus on renewable energy. Since then, the wind power industry has expanded swiftly, even with still relatively low fossil fuel prices[1].

1.1.2 Current State of The Wind Energy Market and Trends

In order to comprehend this rapid change International Energy Agency (IEA) made a prediction that total installed capacity in wind industry will reach 2016 GW by 2050, accounting for over 12% of global electricity consumption [3], in comparison this number in 2023 amounted up to just over 1000 GW (see below 1.1).

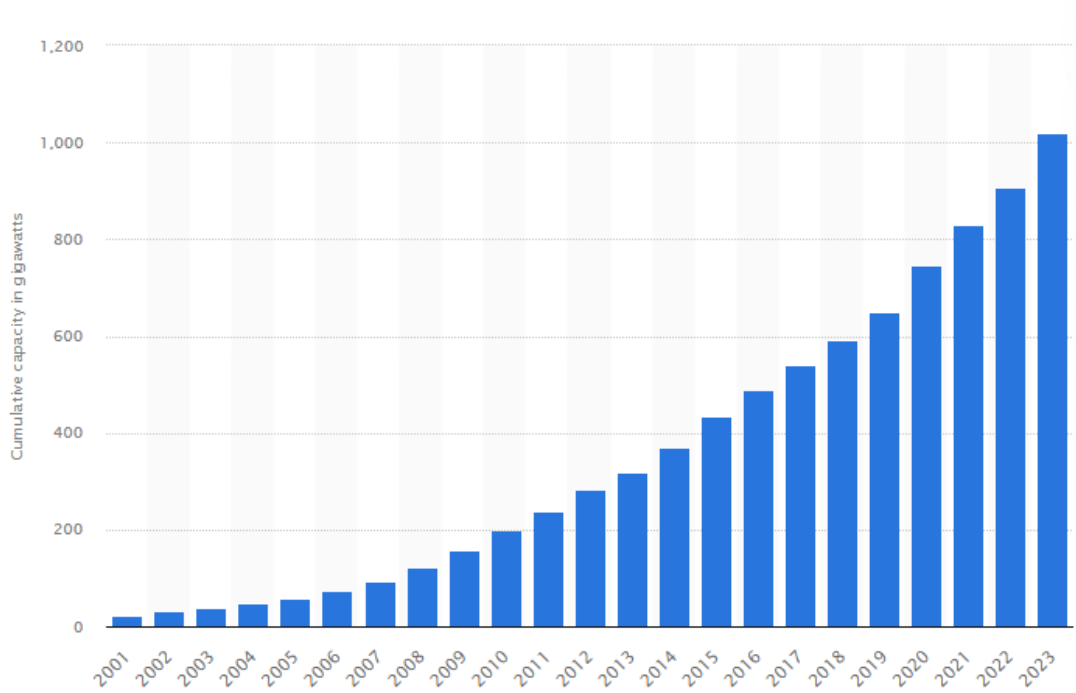


Figure 1.1: Global cumulative installed wind capacity 2001-2023 [4]

More importantly, in 2023, wind energy contributed to a share in global electricity generation of about 8% [5].

On the other hand, despite growing awareness, carbon emissions measured between 2019 and 2023 continued to rise at an average yearly growth rate of 0.8% [6]. Should CO_2 emissions remain near current level, the goal outlined by the Intergovernmental Panel on Climate Change (IPCC) — which aims to limit the probability of global temperature increases to $2^\circ C$ — is projected to be depleted by the early 2040s [6].

In the light of the previous trends, the annual Energy Technology Perspectives (ETP) report published by International Energy Agencies approximates that, if current practices continue without the adoption of new technologies, greenhouse gas emissions from the energy sector could rise by 130% above 2005 levels by 2050 [7].

Addressing these emissions will require an energy technology revolution that encompasses increased energy efficiency, expansion of renewable energy sources, nuclear power utilization and decarbonization of fossil-fuel-based power generation [8]. In order to understand the sustainability implications of transition to wind energy, the above mentioned ETP's report suggests that wind power alone could provide up to 12% of the emission reductions needed within the power sector [8, 7], as is shown in the picture 2.1 below.

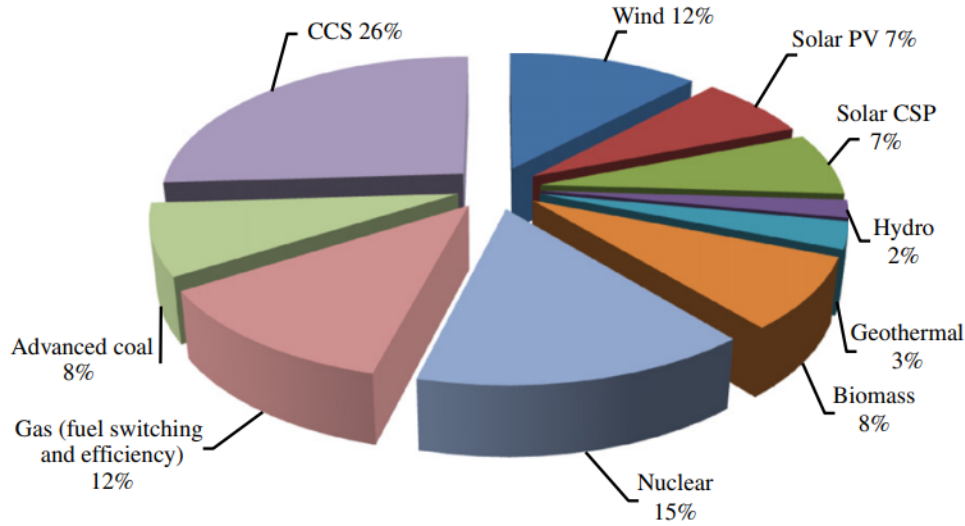


Figure 1.2: Potential contributions of different energy sectors to CO_2 reductions [9]

Beyond the reduction in CO_2 emissions, wind energy can also contribute to reducing pollutants such as sulfur and nitrogen oxides [10, 11].

Moreover, the development of aerodynamics, structural dynamics, and micrometeorology has succeeded in raising turbine energy output annually by 5% [12, 13]. As will be stated, because global energy demand, measured from 2019 to 2023, keeps rising yearly by about 1% [6].

In the meantime, modern wind turbines have become much lighter and quieter with important reductions both in weight and noise achieved within the recent years [8]. In order to understand the nature of this rapid change, when taking into account only past three decades, turbine size has increased by a multiplier of 100, while energy costs have dropped by more than fivefold [8].

Despite all of this, in its present form, the wind industry also remains faced with the challenge of designing highly efficient turbines with proven capacity for converting wind energy into electricity, besides being sustainable on grounds of environmental and economic viability considerations [8].

In order to achieve this, further estimations show, that further increases in wind energy production are possible by strategically selecting wind farm locations, using high-capacity

generators, implementing faster computer-assisted manufacturing techniques, enhancing power factors, and adopting government policies [14, 15]. This implies that advances in wind turbine technology will need to particularly focus on improving efficiency, durability, sustainability and cost-effectiveness [16, 17, 18].

1.1.3 Research Questions

As discussed in previous subsections, the wind energy sector has experienced rapid and extensive growth. At the same time, recently developed sustainability regulations have been introduced and ambitious goals in terms of sustainability have been set for the upcoming decades. These trends have given rise to the need for newer methods and other processes to the industry for fulfilling regulatory requirements and sustainability goals efficiently. For this reason, most companies in the wind energy industry have implemented different initiatives such as carbon neutrality and zero-waste management.

Hence, the primary focus of the thesis is to discuss the different challenges associated with the sustainability goals at every stage of the lifecycle related to wind turbines and by considering Key performance indicators (KPIs), both financial and sustainability ones. The thesis also focuses on understanding the financial and sustainability implications of such measures for sustainability in light of alignment with future policy and industry standards.

On that account, in order to guide this assessment, the research addresses the following main research question:

1. **What are the stages of the wind turbine life cycle and what are the challenges to achieving sustainability goals across the wind turbine life cycle?**

with a following two sub-questions:

1. **Which KPIs are used to measure financial and sustainability impact throughout the life cycle of wind turbines?**
2. **What is the relationship between financial and sustainability KPI trade-offs?**

1.2 Research Methodology

This thesis utilizes qualitative approach, integrating a thorough review of existing literature and analysis insights gathered from the industry experts.

Sources for the literature review include different sources, such as reports from leading industry stakeholders, governmental policy documents and scientific papers on the wind turbines and state of the wind industry. The main objective for the literature review was to assert the phases within the life cycle of the wind turbines and to identify most important Key performance indicators (in the rest of the thesis will be addressed as *KPIs*) which are tracked, as well, where possible, identify the specific sustainability challenges and improvement opportunities associated with each stage.

Besides this literature review, a interview with sustainability expert of leading wind energy sector company was conducted. Goal of interviews is to present the industrial view of current gaps in sustainability throughout the life cycle and what might be the potential steps to prevent them, as well as to identify which sustainability KPIs are considered most critical. Further, the interview shows how trade-offs between sustainability and financial considerations are performed, hence offering a view on how the industry navigates this relationship.

1.3 Structure of the Thesis

The thesis is structured as follows:

- *Introduction* – The problem statement in terms of defining objectives, questions, methodology, and layout of the research work.
- *Wind Turbine Life Cycle Stages* – Defining different stages that make up a life cycle of wind turbines by providing an in-depth overview to form a basis for further discussions and discusses current challenges towards the attainment of sustainability goals.
- *Tracked Financial and Sustainability KPIs* – Provides key performance indications that are tracked throughout the life cycle and goes on to do in-depth analysis
- *Financial and Sustainability KPI Trade-off* – Explores the interplay that would exist between financial and sustainability KPIs. It looks at the trade-offs that would be necessary when balancing economic objectives against environmental ones.
- *Conclusion* – A summary of findings, implication of findings, and any possible directions for a sustainable future in wind energy.
- *Bibliography* – A complete list of all references quoted in the thesis.
- *Appendix* - Contains the interview guideline with interview questions and full transcript of interview with the industry expert.

Wind Turbine Life Cycle Stages

In order to assess and understand the environmental and economic sustainability impact of wind turbines, it is crucial to understand the distinct phases within their life cycle.

2.1 Classification of Life Cycle Stages

2.1.1 Variability in Life Cycle Stages

While numerous studies, papers and reports have addressed the life cycle of wind turbines to different extents, there is no universally agreed-upon division of stages. Sources differ in how they categorize the phases based on specific focus areas, such as environmental impact, process complexity, type of the turbine (offshore vs. onshore turbines) or economic factors. Consequently, some papers propose fewer, broader stages, while others introduce division with more detailed substages. This lack of standardization makes direct comparisons across studies very difficult, since different categorizations may indicate variations in reported impacts.

2.1.2 The Four-Stage Model

However, despite the variations mentioned above, it could be stated that overall there is a broad accepted consensus in the wind turbine literature in this regard (with slight differences). As introduced in the papers [19, 20, 21], which were fundamental for the life cycle analysis conducted in this thesis and in the context of examining the life cycle on multiple different layers (environmental and financial) the rest of the thesis rests on the fact, that life cycle of wind turbines comprises of following four stages:

- Extraction and Processing of Raw Materials,
- Manufacturing and Installation,
- Operation and Maintenance,
- Decommissioning and Recycling

It is important to mention that in some sources, such as [19, 21] the transportation is considered as a separate stage. Instead, in this thesis transportation will be observed as an integral part of each of the above. This is in agreement with the methodology in [20], in which transportation involved in each stage is considered within that stage to avoid double counting and show its contribution in each life cycle stage.

This standardized four-stage approach aligns well with the requirements of a Life Cycle Assessment (LCA), which was used in the above mentioned literature and which seeks to evaluate environmental impacts throughout the entire life span of wind turbines [20, 19]. Furthermore, this approach also reflects key differences between onshore and offshore turbines, material needs and maintenance logistics vary significantly, though each stage remains essential to a complete sustainability analysis [19].

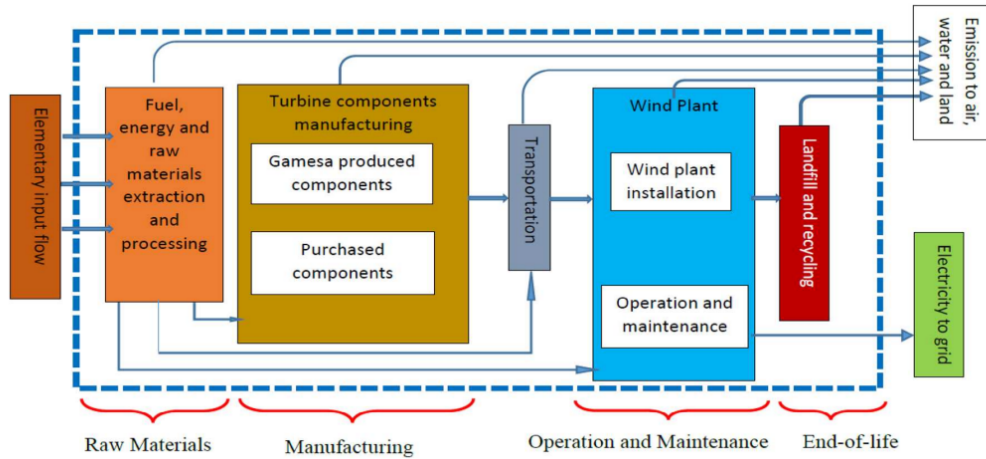


Figure 2.1: Wind turbine life cycle stages on the example of Gamesa wind turbine [21, 22]

2.2 Detailed Examination of Life Cycle Stages and Sustainability Challenges

2.2.1 Extraction and Processing of Raw Materials

This stage typically involves extracting the different metals, such as, iron, copper, aluminum and so on, as well as obtaining materials like steel, concrete, fiberglass, and rare earth elements, which are essential for constructing main components of wind turbines such as the tower, nacelle, rotor blades, gearbox and generator [20, 23, 24, 25]. The major environmental impact in this stage relates to the high energy requirements needed for mining, refining, and processing, which in turn, contribute to very significant greenhouse gas emissions [26, 27, 25].

Steel is the primary input in manufacturing turbine towers, that accounts for a large share of the material weight [21, 26, 28]. Studies indicate that aside from steel production, which is one of the most resource-intensive processes in this phase, with substantial amounts of fossil fuels consumed, also the production of concrete for foundations adds to the environmental load, especially in onshore installations where larger foundations are typically required [21, 27, 19].

The extraction and use of rare earth elements, particularly neodymium, which is used in permanent magnets as a part of turbine generators, present further environmental challenges. Such elements as neodymium are primarily mined and trade dominated in a few global regions (such as China), often under conditions that lack stringent environmental regulations, leading to local ecological degradation and water contamination. Furthermore, processing of rare earth elements in itself is a long and energy-intensive process, through which the elements from the ore are separated through successive steps, which in turn create large amounts of waste and toxic by-products [26].

Transportation logistics for raw materials to manufacturing facilities also play an important role in the environmental impacts of this stage. Studies covering installations in Europe and the United States underline that transportation distances and different modes can largely influence emissions. Taking an example, transportation by ship and truck over long distances, especially for offshore installations, increases the carbon footprint of this stage significantly [19, 21].

Hence, this stage provides the initial basis for any improvements in terms of sustainability of the wind turbine life cycle (in this, as well as in other stages, such as recycling stage), by e.g exploring and adopting the alternative materials and more efficient processing techniques [23, 28, 25].

2.2.2 Manufacturing and Installation

The Manufacturing and Installation phase of wind turbines includes the production of main turbine components (such as the tower, nacelle, rotor blades and generator), as well as transporting and piecing together together these parts at the installation site [24, 27, 25, 28].

The manufacturing of turbine towers, which are mostly made from steel, is one of the most resource-demanding in this processes during this stage. Steel production alone accounts for a significant share in both energy consumption and carbon emissions during the manufacturing stage of the turbine, especially in combination with onshore turbines with concrete foundations. Some studies estimated that more than 50% of the manufacturing environmental impact of towers and foundations is constituted by steel and concrete, especially because of the high level of emissions related to the production of steel [27, 20, 28, 21].

Furthermore, the primary construction materials for rotor blades include fiberglass and other composite materials. Despite their specific advantages with respect to durability and performance, they are fairly difficult to produce sustainably, as production processes tend to be highly energy-intensive, using considerable amounts of fossil fuel for the materials in question, along with very limited recycling options. Moreover, rotor blade manufacturing has been known for its high consumption of synthetic resins and other polymers from non-renewable resources that are generally poorly recyclable, raising concerns over sustainability issues [19, 21, 24, 25].

On the other hand, this phase contributes to environmental impact due to transportation of the components to the installation site, especially in case of offshore wind farms. Transportation of large turbine components over long distances, for example, involves heavy ship and heavy trucks consuming large amounts of fuel, hence resulting in greenhouse gas emissions. In the case of offshore installation, it may require more equipment and even vessels specially adapted for carrying and laying the components, increasing the energy demand and the related emission in this stage. [28, 20, 21, 24].

Other relevant activities involved in this phase are installation activities: foundation laying, tower hoisting and turbine component assembly [27, 21]. The onshore turbines have large concrete foundations, while the offshore ones rely on steel-driven monopiles or other support structures, depending on the depth of the sea and conditions of the seabed [19, 25]. This installation phase is not only material-intensive but also highly energy-consuming by the construction machinery and transport, especially in offshore sites where heavy lifting and maritime building equipment are required [27, 28].

2.2.3 Operation and Maintenance

The operation and maintenance phase of wind turbines, which spans most of the turbine's life, includes activities that enforce continuity in the functionality and efficiency of the turbines. In comparison, this stage generates minimal direct emissions since the generation of wind energy does not emit any greenhouse gases. However, from an operational point of view, periodic maintenance, replacement of parts and monitoring of the systems still impose an environmental burden [27, 25, 21, 28, 20, 26].

Routine maintenance includes the periodic inspections, lubrication, and repairs needed to ensure that failure is avoided and optimal performance maintained [25, 21]. The frequency of the applied maintenance schedule varies with both the type and the site of the turbine, whereas the offshore turbines require more intensive and more frequent maintenance due to harsh weather and location conditions [19, 27, 25]. While essential in themselves, these activities contribute to transportation, energy, and resource use which adds to the overall impact in this stage [27, 21].

Another significant element of this operation and maintenance phase is the replacement of parts [27, 21]. A lot of mechanical parts, such as bearings, gearboxes and hydraulic systems, wear out during the life span of a turbine and may need to be changed on multiple occasions [27, 21, 28, 20]. This includes the manufacturing and transportation of new parts, which reintroduces some environmental cost from the manufacturing stage [27, 21]. Several studies indicate that gearless turbines usually require less maintenance, because they lack some of the components which are subject to wear out in geared turbines [27, 26].

For offshore wind farms, maintenance processes raise special logistical and environmental challenges [19, 20]. Regular access to such sites (especially in case they are remote) can require specialized vessels, which consume fossil fuel and generate emissions [19, 21]. The schedule for maintenance, based on weather conditions, can also complicate further and lead to longer downtimes and more emissions due to extra standby periods of equipment [19, 20, 26].

In general, during the operation and maintenance phase lower emissions are released compared to other stages. However, it is still a critical phase required for longevity and efficiency of the turbine. Therefore, proper handling of the activities involved during maintenance would enable performance optimization and minimize environmental impacts, thereby aligning with long-term sustainability goals [27, 25, 21, 28, 20, 26].

2.2.4 Decommissioning and Recycling

The Decommissioning and Recycling phase presents the end of the life cycle, with main activities including the dismantling, transport and processing of the turbine parts in order for them to be prepared for disposal or recycling. This process involves substantial logistical preparations that have to be done in order to minimize environmental impacts, especially given the size and complex structures of towers, blades and foundations. [19, 27, 26].

Major components of wind turbines, such as the tower and nacelle have to be dismantled. For this purpose heavy machinery is needed. Components are either taken directly to recycling facilities upon dismantling or taken to other sites for staging and further processing. The tower and other parts made from steel usually have high rates of recyclability, given the value of the steel and its potential for reuse [26, 28, 19, 27, 25].

Rotor blades are to some extent different when discussing decommissioning stage. Being primarily manufactured from composite materials such as fiberglass and resin, the recycling of blades produces operational challenges regarding the separation and reuse of their materials. Hence, at this stage blades are often landfilled or incinerated, which can have negative environmental impacts [27, 25, 20].

The foundation is another large component, notably for onshore turbines with large concrete bases and for offshore turbines with submerged structure components. Most often, decommissioning of the foundations leaves portions of the structure in place, if complete removal would be disruptive to the ecology. Furthermore, offshore foundations (e.g. monopiles) might be partly decommissioned up to beneath seabed levels in order to reduce marine impact [19, 27, 28, 26].

Transportation of the decommissioned materials recycling or disposal facilities further contributes to the environmental load at this stage. Since the size and weight of the turbine parts are quite big, the related emissions due to transportation can be substantially high, especially if recycling facilities are located far from the installation site [28, 21, 27, 19].

Tracked Financial and Sustainability KPIs

In this chapter the most important tracked financial and sustainability KPIs will be discussed, based on the literature findings.

3.1 Sustainability KPIs

3.1.1 Carbon Footprint (CF)

The carbon footprint KPI measures the volume of greenhouse gases that are produced and emitted throughout a wind turbine's life cycle, from extraction and processing of raw materials to decommissioning and recycling. This KPI is essential in providing the wind turbine's environmental performance and informs the stakeholders about its contribution toward the reduction of carbon emissions in comparison to other fossil fuel sources of energy. Through CF, wind farm operators are able to benchmark their performances against industry thresholds and set greenhouse gas emission reduction targets [29].

3.1.2 Energy Return on Investment (EROI)

EROI is one of the major KPIs for sustainability. Briefly, it calculates the ratio of energy generated through the wind turbine to the amount that has to be invested in manufacturing the turbine, operating it, and disposing of it at the end. The higher the EROI, the more sustainable the source of energy, since the turbine produces much more energy than will ever be consumed in the turbine's life cycle. This metric can be very useful in comparing basic renewable energy sources to with traditional fossil fuels ones [29, 30].

3.1.3 Waste Generation and Material Recyclability

This KPI measures waste generated within a wind turbine life cycle and the percentage of materials that could be recycled. Wind turbines are composed of elements with different levels of recyclability: metal, composites, and lubricants. Waste and recyclability monitoring offer insight into the real environmental sustainability of the used materials and encourages the adoption of more sustainable materials and processes [29].

3.1.4 Water Usage

Water usage tracks the volume of water needed throughout the life cycle of wind turbines, including manufacturing, installation and operation. Taking into account that water resources are limited, its conservation is essential for sustainability. This KPI is particularly important to arid regions where water is scarce, hence enhancing resource efficiency. [29].

3.1.5 Land Use Efficiency

This KPI evaluates how well the area occupied by wind farms is used, in relation to its environmental impact. Land use takes into account various factors, such as turbine spacing, the possibility of land restoration after decommissioning and any other impacts on local ecosystems. A higher efficiency means low environmental footprint against energy produced [30].

3.1.6 Total Resource Use

Total resource use takes into account multiple resource inputs, such as metals, energy, and water, required throughout the turbine's life cycle. This KPI is monitored to ensure resource usage is in line with sustainability objectives through providing potentialities for reduction of material input, gains in efficiency or substituting with more viable materials. [29].

3.2 Financial KPIs

3.2.1 Operational Expenditures (OPEX)

OPEX refers to the ongoing costs required for the wind farm to operate. It encompasses regular operation, maintenance and unscheduled repair activities. This is one of the most important KPIs because a big portion of the budget goes to this area and it is tracked so that the financial viability of the wind operations can be realized. It should be normalized by capacity to enable comparability [30].

3.2.2 Earnings Before Interest, Taxes, Depreciation, and Amortization (EBITDA) Margin

EBITDA margin reflects the proportion of revenue remaining after accounting for operating expenses. It can be used to conduct the comparison in profitability across different wind farms. It is important to emphasize, that EBITDA does not take into account any capital expenditures, taxes, or interest that might be influencing liquidity and all of financial health. [30].

3.2.3 Loan Life Coverage Ratio (LLCR)

LLCR presents the ratio of the net present value of cash flows to outstanding debt and reflects thereby the capability of the wind farm to repay its long-term debt. Therefore, this KPI provides insights into the financial stability of the project during its lifetime and thus is relevant for both investors and operators [30].

3.2.4 Debt-Service Coverage Ratio (DSCR)

DSCR refers to a wind farm's ability to pay off its annual debt, by relating available cash to the debt. This is generally used for understanding the potential to repay short-term debt and supports a snapshot of annual financial health [30].

3.2.5 Free Cash Flow to Equity (FCFE)

FCFE is defined as the cash earned by a company after accounting for all expenses, debt payments, and reinvestments.. This is an important KPI from an investor point of view, in terms of profitability [30].

3.2.6 Levelized Cost of Energy (LCOE)

LCOE calculates the total cost per unit of energy over the wind farm's life cycle. It is used to compare economic efficiency across energy sources. This KPI provides a standardized view of the cost of energy, though usually benchmarked against installed capacity rather than energy output [30, 31].

3.2.7 Break-even Price of Energy (BEPE)

BEPE represents the selling price of energy that a wind farm needs to break even. Therefore, it offers a realistic view of profitability under varying economic conditions. It is used for profitability analysis, especially in cases where the inflation rate and tax rates are different [30].

Financial and Sustainability KPI Trade-off

This section analyzes the trade-offs encountered between financial and sustainability KPIs across the wind turbine life cycle, highlighting the complexities of achieving both low costs and minimal environmental impacts. The literature content on this topic will be synthesized and the findings of interview with industry expert will be discussed. Lastly, the comparison between the two will be made.

However, it is important to emphasize that there is a shortage of papers which cover the trade-offs which are discussed here. Only few papers, some even partially, discuss to some extent given problematic. Therefore, for this part of literature review the synthesis of two different relevant scientific papers will be used.

4.1 Literature Findings

In the operational phase, the greenhouse gas emissions from wind turbines are lower compared to the rest of the energy industry, hence, the role of the wind sector will be very important in renewable energy and decarbonization efforts [20].

However, in its life cycle, it has significant environmental costs, especially during the construction phase. The material extraction processes that involve the manufacture of steel and concrete for towers contribute heavily to global warming potential (GWP), acidification, and resource depletion. In addition, rare earth metals, neodymium in particular, used in turbine magnets, further increase the environmental impacts caused by energy-intensive extraction processes with scarce recycling possibilities [20].

Similarly, wind power plant (WPP) construction as a consequence disrupts ecosystems through exploitation and long-term change of land, carbon emissions from deforestation,

and the loss of natural carbon sinks like forests. This is particularly evident when WPPs are placed in biodiversity-rich or wilderness areas, whose purpose is to protect critical natural capital. These negative impacts present real environmental trade-offs arising as a consequence of the expansion of wind energy infrastructure [32].

Therefore, biodiversity and wilderness constraints make fewer sites viable to install WPPs, thereby shifting projects to more populated areas. These actions preserve ecological integrity in untouched areas, but on other hand on the cost of noise, visual disturbances and reduced household welfare [32].

This means that more profit would have to be sacrificed in terms of system surplus (ESS) cost and investment in less profitable locations that have lower wind efficiency or higher grid connection costs. sing Norway as an example, southern regions provide the highest ESS due to superior wind conditions and a higher price of electricity. On the other hand, biodiversity constraints often force investments northward where ecological impacts are less but financial returns are diminished [32].

Sustainability challenges also remain during the decommissioning stage. At the end of a turbine's life, on one hand recyclable materials such as steel can reduce environmental impacts by up almost a half, composite materials used in rotor blades are often landfilled or incinerated, contributing to increasing environmental problems. To reduce end-of-life impacts, significant financial investments in recycling technologies are needed [20].

Both papers find solutions that can minimize these trade-offs. For example, larger turbines decrease the environmental impacts associated with per unit energy generation and improve life cycle cost efficiency, even though upfront capital costs are higher [20].

Similarly, monetized environmental and disadvantage costs integrated into energy system models support policymakers in finding the optimal configurations to maximize financial benefits while preserving the ecology [32].

Recycling technologies for composite materials and innovative low-emission steel production represent promising technologies. However, all these solutions require significant financial commitments and supportive policy frameworks in order to become profile as a standard within the market [20].

Besides, policies incentivizing non-price criteria in WPP auctions, such as biodiversity protection, could be a good measure, could encourage the industry players to take into account more sustainable siting and operational practices [32].

4.2 Interview Insights

The interview was conducted with an industry expert, Andrew Hardwell (for the full interview transcript see the Appendix). Andrew works in a sustainability team of Vestas, one of the leading wind energy companies and is stationed in Copenhagen office. This current role within sustainability department is the Sustainability Specialist. He has substantive expertise and background in ecology and environmental sociology, and his

primary roles involve the development of sustainable products and helping to shape the sustainability strategy for the company. The discussion revolved around the important challenges in sustainability, financial trade-offs, important KPIs, and industry practices correlated with the life cycle of wind turbines.

According to him, supply chain emissions reduction- particularly material value chains like steel, iron, copper, and aluminum-decarbonization-remains one of the most urgent challenges that the wind energy sector is facing. He emphasized that marine gas fuels used for long-distance transportation and even in logistics services in offshore wind farms have a big environmental impact. Further, he pointed out that critical issues included the lack of recycling solutions for composite materials, such as in wind turbine blades, and the possible adverse impacts on avian fauna. On life cycle stages, Andrew remarked that the most significant sustainability issues occur in the upstream stages, in particular raw material mining and refining, as these have the most significant environmental impact.

The sustainability and financial objectives of the wind energy sector represent a combination of alignment and tension. He explained that sustainability initiatives can reduce exposure to carbon trading schemes and tariffs, improving project competitiveness. He provided examples of non-price criteria, e.g. protection of biodiversity and reduction of emissions, that influenced the awarding of offshore wind project auctions in Europe. He also pointed out that decarbonization of the logistics offshore and switch to renewable or bio fuels for all road transport, vessels, and facilities usually increase the costs, often critical in view of financial feasibility.

From the financial point of view, the construction and service phases of wind turbines were the most resource-intensive due to high emissions and associated costs. For wind energy as a whole, the upstream levels of extraction and refinement of materials were again addressed as the most costly and impactful. Innovative practices for these challenges involve using low-emission steel manufactured from recycled scrap powered by renewable energy; such steel has emissions decreased by a half. Future developments include near-zero emission steel, he said, using hydrogen instead of coal for the reduction process. Speaking on circularity, Andrew said that one of the ongoing projects, entailing the recycling of composite materials like fiberglass and carbon fiber, which are currently difficult to recycle, is a major step toward a circular economy.

Among the most critical KPIs on sustainability, one can name scope 1, 2, and 3 emissions and material efficiency, tracking waste creation and recycling rates.

Small digression, for the context the terms of scope 1, 2, and 3 emission will be explained. The Scope 1 emissions include direct emissions from sources owned or controlled by an organization, involving activities such as the burning of fuel in company vehicles, provided they are not electrically powered. Scope 2 emissions involve indirect emissions that occur due to production of the energy purchased and used by a company. Scope 3 emissions come from activities throughout the company's value chain related to the manufacturing process, transportation, usage, and disposal of products from its suppliers. The emission sources are not under direct possession or control of the organization but

are influenced by its functions [33, 34].

Back to the interview, from a financial standpoint, it was underlined that the primary KPIs needed to be directed toward reducing exposure to carbon-related taxes and tariffs and improving project margins through sustainable offerings. He pointed out that stakeholders, regulators and investors shape the focus on sustainability through such mechanisms as carbon trading schemes and non-price criteria in project bids, thereby fostering alignment of the industry's financial objectives with those of sustainability.

He argued that financial sustainability is a prerequisite for any project, wherein economic viability would ensure the continuance of business operations. At the same time, he cited the importance of thresholds to avoid projects which could present environmental or social risk, even if lucrative from a financial point of view. Offshore wind turbines, on the other hand, present more sustainability challenges than onshore turbines, mainly due to the difficulty of decarbonizing logistics at sea. In the longer term, he forecasted more disclosure of environmental, social, and governance (ESG) metrics and their active use strategically to ensure competitiveness and cost-efficiency within the wind sector. He also raised a provocative question about whether prioritizing cost reduction in wind energy, even at the expense of additional sustainability measures, may accelerate the transition away from fossil fuels.

4.3 Comparison Between Literature and Industry Praxis

4.3.1 Similarities

One of the notable similarities is the emphasis on the raw material extraction and processing phase as a critical life cycle stage. In both cases, the literature and the interview identify this very stage as resource-intensive and most impactful towards environmental sustainability.

The literature points out the substantial emissions which arise during steel and concrete production, as well as the degradation of the environment due to mining rare earth elements such as neodymium. This concern is reflected in the interview, where the expert mentioned that supply chain emissions, which are product of steel, aluminum, and copper production, present one of the areas which need to be a subject of further and detailed discussions.

Another such alignment is the recognition of the limitation in the recycling of composites, particularly turbine blades. The literature and the industry perspective both put great emphasis on the difficulty in recycling fiberglass among other composites, which often end in landfills or are incinerated. In this respect, the shared focus on innovation in recycling technologies as the route to sustainability becomes a key component in achieving sustainability goals.

It is agreed on both sides that decarbonizing operations is a long-term objective. The literature discusses the environmental benefits that come with reduced emissions through

advanced manufacturing processes and improved maintenance strategies, while the interview identified other practical steps which could be taken, such as transitioning to low-emission steel and the development of circular economy models for composite materials.

4.3.2 Differences

While some overlapping takes place, there are striking differences between the theoretical perspective and practice, especially with regard to the ranking of KPIs. The literature extensively discusses financial KPIs such as EBITDA, OPEX, and LCOE (see chapter 3.1. for more information), which are said to be critical for a project's economic viability. These metrics are put as central in balancing financial objectives with sustainability considerations, for which the most relevant KPIs include CF, EROI and others (see chapter 3.2. for more information). However, what the interview reveals is different from this regard. While financial KPIs remain important, however now with emphasis on reducing exposure to carbon related taxes and tariffs. In terms of sustainability, the expert pointed to scope 1, 2, and 3 emissions and material efficiency as the most critical sustainability KPIs in terms of sustainability. These metrics are more aligned with long-term environmental goals and increasingly influenced by stakeholder pressures, such as regulations and carbon trading schemes.

Another key difference is related to perceived role of sustainability trade-offs. The literature often frames such trade-offs as necessary compromises, especially in stages like manufacturing, where material costs and emissions are closely intertwined. In contrast, this interview underlines the possibility of interlinking financial and sustainability goals—for example, low-emission steel would reduce environmental impact while at the same time improving the competitive positioning in bids for projects. From this perspective, innovation and stakeholder demands would tend to be viewed more optimistically in aligning these goals in practice.

Additionally, the financial implication of sustainability remains generally a second concern in most literature, which focuses on the environmental side of KPIs. However, the interview shows the practical relevance of carbon taxes and tariffs as financial levers shaping sustainability strategies. The practical focus here indicates an integrated approach—both financial and environmental factors must be addressed together. With this in mind, interview emphasizes the importance of economical KPIs and economic viability, without which no sustainability goals could be achieved.

Lastly, the mentioned similarities and differences will be summarized and shown in a compact table, see 4.2 below:

Aspect	Literature Insights	Interview Insights	Comparison
Raw Material Extraction and Processing	Identified as the most resource-intensive phase with substantial emissions from steel, concrete, and rare earth mining.	Raw material extraction was emphasized as critical. Industry expert highlighted supply chain emissions from materials like steel, aluminum and copper.	Aligned
Recycling Limitations	Stressed challenges in recycling composite materials, particularly turbine blades. Materials often end up in landfills or are incinerated.	Similar concern about recycling composites. Expert emphasized need for innovation in recycling technologies.	Aligned
Decarbonization	Discussed environmental benefits of reducing emissions, by using advanced manufacturing and maintenance strategies.	Supported decarbonization with practical steps, such as transitioning to low-emission steel and developing circular economy models for composite materials.	Aligned
Financial KPIs	Financial KPIs like EBITDA, OPEX, and LCOE are emphasized as key for evaluating economic viability and balancing financial and sustainability goals.	Financial KPIs remain important, with emphasis however on exposure to carbon related taxes and tariffs	Opposed
Sustainability KPIs	Focus on environmental KPIs such as emissions reduction, energy efficiency, and material efficiency.	Focus on emissions (Scope 1, 2, 3) and material efficiency as the most critical KPIs in practice.	Aligned

Aspect	Literature Insights	Interview Insights	Comparison
Trade-offs	Mentioned trade-offs as necessary compromises, particularly in stages like manufacturing, where material costs and emissions are closely linked.	Emphasized potential alignment of financial and sustainability goals, e.g., using low-emission steel to reduce environmental impact and improve competitiveness.	Opposed
Finance vs. Sustainability	Financial concerns often treated as secondary, focusing primarily on environmental impacts.	Highlighted the importance of carbon taxes and tariffs, suggesting that financial factors and sustainability strategies are increasingly intertwined, with emphasis on financial side.	Opposed

Table 4.2: Comparison of Literature and Interview Insights on Sustainability Challenges and KPIs in Wind Turbine Life Cycles

Conclusion

This thesis systematically investigated the sustainability challenges and trade-offs in the life cycle of wind turbines, covering each life cycle stage in view of the financial and sustainability KPIs involved, and discussing implications of aligning theoretical insights with practical challenges.

The analysis identified four stages in the life cycle of wind turbines, each having different sustainability challenges:

- *Extraction and Processing of Raw Materials* - while the actual processing of steel, concrete and rare earth elements is highly greenhouse gas-emission-intensive, rare earth mining has a number of serious social and environmental impacts, making the sourcing of these materials quite complex.
- *Manufacturing and Installation* - this stage is energy-intensive, with emissions stemming from material production and logistics. In particular, it was mentioned that offshore installations are difficult because of transportation and requirements for special equipment.
- *Operation and Maintenance* - although less emission-intensive, this phase also entails frequent maintenance activities often with emissions related to transportation and replacement of components, especially for offshore projects.
- *Decommissioning and Recycling* - recycling of the steel components is common, while composite materials, such as rotor blades, still have unsolved recycling problems. While there is a potential for recycling, this is also limited by current technological capabilities.

The critical sustainability KPIs included inter alia Carbon Footprint, Energy Return on Investment, and Material Recyclability, which are necessary for the identification and evaluation of environmental impacts in every stage of the life cycle of the turbine.

Financial KPIs included inter alia LCOE, OPEX, and DSCR shed light on the economical factors affecting project feasibility.

Among these KPIs, the tension between sustainability achievement and cost-effectiveness is underlined in high-emission material procurement stages. In the raw material and manufacturing stages, significant trade-offs are implied, as a reduction in emissions often leads to higher material costs (e.g. near-zero-emission steel)

Regulatory mechanisms, like carbon tax, along with the market forces of non-price criteria in auctions, form the influencing capabilities of stakeholders that help balance financial and environmental goals.

Both literature and interview findings pointed out that raw material extraction is the most resource-intensive stage, hence requiring low-emission alternatives. Although literature shows sustainability to be a financial trade-off, the industry experts pinpointed possible opportunities for alignment, such as regulatory incentives and market-driven initiatives on sustainability. Industry insights showed an increasing focus on composite material recycling and maintenance logistics, though with evident limitations from technology and cost.

This study concludes that wind turbines are important in renewable energy transitions but pose significant sustainability challenges throughout their life cycle, which are not usually talked about. Such challenges could be mitigated through innovation integration, regulatory support, and improved recycling competencies within the sector. The better linkage of financial and sustainability KPIs, therefore, together with stakeholder cooperation, allows for more environmentally friendly wind energy systems. Therefore, one main takeaway is that the financial and sustainability side are intertwined and would be hard to discuss, without taking into account other side.

Future research, which has broader goal of achieving a net-zero energy sector, should focus on developing advanced recycling methods for composite materials, exploring low-emission materials and using emission-free vehicles.

Bibliography

- [1] Qiuwei Wu and Yuanzhang Sun. *Modeling and modern control of wind power*. John Wiley & Sons, 2018.
- [2] CF Brush. Mr. brush’s windmill dynamo. *Scientific American*, 63(25):389–389, 1890.
- [3] IEA WIND Energy and A Iea. Iea. *Renewable Energy Market Update*, 2020.
- [4] Statista Research Department. Cumulative installed wind power capacity worldwide from 2001 to 2023, 2024. Accessed: 2024-11-06.
- [5] Statista. Share of electricity generation from wind energy sources worldwide from 2010 to 2023, 2023. Accessed: 2024-11-06.
- [6] BP Energy Outlook 2024, 2024. Accessed: 2024-11-06.
- [7] International Energy Agency. *Energy Technology Perspectives 2006*. 2006.
- [8] Mohammad Rafiqul Islam, Saad Mekhilef, and Rahman Saidur. Progress and recent trends of wind energy technology. *Renewable and Sustainable Energy Reviews*, 21:456–468, 2013.
- [9] Global Wind Energy Council. Global wind statistics 2012. *Global Wind Report*, 2013.
- [10] NA Madloul, Rahman Saidur, NA Rahim, MR Islam, and MS Hossian. An exergy analysis for cement industries: an overview. *Renewable and Sustainable Energy Reviews*, 16(1):921–932, 2012.
- [11] Tugrul U Daim, Muhammad Amer, and Rubyna Brenden. Technology roadmapping for wind energy: case of the pacific northwest. *Journal of Cleaner Production*, 20(1):27–37, 2012.
- [12] Oscar Barambones, Jose Maria Gonzalez de Durana, and Enrique Kremers. A neural network based wind speed estimator for a wind turbine control. In *Melecon 2010-2010 15th IEEE Mediterranean Electrotechnical Conference*, pages 1383–1388. IEEE, 2010.

-
- [13] Z Hameed, YS Hong, YM Cho, SH Ahn, and CK Song. Condition monitoring and fault detection of wind turbines and related algorithms: A review. *Renewable and Sustainable energy reviews*, 13(1):1–39, 2009.
 - [14] M Islam, N Rahim, K Solangi, and Rahman Saidur. Assessing wind energy potentiality for selected sites in malaysia. *Energy Education Science and Technology Part A-Energy Science and Research*, 29(1):611–626, 2012.
 - [15] PJ Musgrove. Wind energy conversion: recent progress and future prospects. *Solar & wind technology*, 4(1):37–49, 1987.
 - [16] Ali Mostafaeipour. Productivity and development issues of global wind turbine industry. *Renewable and Sustainable Energy Reviews*, 14(3):1048–1058, 2010.
 - [17] Ezio Sesto and Claudio Casale. Exploitation of wind as an energy source to meet the world’s electricity demand. *Journal of Wind Engineering and Industrial Aerodynamics*, 74:375–387, 1998.
 - [18] Pavlos S Georgilakis. Technical challenges associated with the integration of wind power into power systems. *Renewable and sustainable energy reviews*, 12(3):852–863, 2008.
 - [19] Alexandra Bonou, Alexis Laurent, and Stig I Olsen. Life cycle assessment of onshore and offshore wind energy-from theory to application. *Applied Energy*, 180:327–337, 2016.
 - [20] Burcin Atilgan Turkmen and Fatos Germirli Babuna. Life cycle environmental impacts of wind turbines: A path to sustainability with challenges. *Sustainability*, 16(13):5365, 2024.
 - [21] Ali Alsaleh and Melanie Sattler. Comprehensive life cycle assessment of large wind turbines in the us. *Clean Technologies and Environmental Policy*, 21:887–903, 2019.
 - [22] Neil D’Souza, Erhi Gbegbaje-Das, and Peter Shonfield. Life cycle assessment of electricity production from a vestas v112 turbine wind plant. *Final report, PE North West Europe ApS, Copenhagen, Denmark*, 2011.
 - [23] Chaouki Ghenai. Life cycle analysis of wind turbine. 2012.
 - [24] Brice Tremeac and Francis Meunier. Life cycle analysis of 4.5 mw and 250 w wind turbines. *Renewable and sustainable energy reviews*, 13(8):2104–2110, 2009.
 - [25] Fulvio Ardente, Marco Beccali, Maurizio Cellura, and Valerio Lo Brano. Energy performances and life cycle assessment of an italian wind farm. *Renewable and Sustainable Energy Reviews*, 12(1):200–217, 2008.
 - [26] Simon Davidsson, Mikael Höök, and Göran Wall. A review of life cycle assessments on wind energy systems. *The International Journal of Life Cycle Assessment*, 17:729–742, 2012.

-
- [27] Begoña Guezuraga, Rudolf Zauner, and Werner Pölz. Life cycle assessment of two different 2 mw class wind turbines. *Renewable Energy*, 37(1):37–44, 2012.
 - [28] Eduardo Martínez, Félix Sanz, Stefano Pellegrini, Emilio Jiménez, and Julio Blanco. Life cycle assessment of a multi-megawatt wind turbine. *Renewable energy*, 34(3):667–673, 2009.
 - [29] Damon Turney and Vasilis Fthenakis. Environmental impacts from the installation and operation of large-scale solar power plants. *Renewable and Sustainable Energy Reviews*, 15(6):3261–3270, 2011.
 - [30] Sebastian Pfaffel, Stefan Faulstich, and Shawn Sheng. Recommended key performance indicators for operational management of wind turbines. In *Journal of Physics: Conference Series*, volume 1356, page 012040. IOP Publishing, 2019.
 - [31] Elena Gonzalez, Emmanouil M Nanos, Helene Seyr, Laura Valdecabres, Nurseda Y Yürüşen, Ursula Smolka, Michael Muskulus, and Julio J Melero. Key performance indicators for wind farm operation and maintenance. *Energy Procedia*, 137:559–570, 2017.
 - [32] Kristine Grimsrud, Cathrine Hagem, Kristina Haaskjold, Henrik Lindhjem, and Megan Nowell. Spatial trade-offs in national land-based wind power production in times of biodiversity and climate crises. *Environmental and Resource Economics*, 87(2):401–436, 2024.
 - [33] McKinsey & Company. What are scope 1, 2, and 3 emissions?, 2024. Accessed: 2024-11-21.
 - [34] National Grid. What are scope 1, 2, and 3 carbon emissions?, 2024. Accessed: 2024-11-21.

Appendix

A.1 Interview Question Template and Guideline

The goal of this interview is to try to explore and compare industry perspectives with the literature findings on the sustainability challenges, critical KPIs and the trade-off between financial and environmental objectives. The interview will be conducted in an open-ended format and divided into three main areas and question within each will be listed. Lastly, the interview will be afterwards transcribed for qualitative analysis.

Part 1: View on Sustainability and Trade-offs

- Could you briefly describe your expertise and role within the wind energy sector?
- In your view, what are the most pressing sustainability challenges in the wind energy sector today?
- Are there specific lifecycle stages where these challenges are most notable?
- How do you perceive the interaction between sustainability and financial objectives in the wind energy sector (are they aligned, interconnected, opposite etc.)?
- Which aspects of wind turbine lifecycles would in your opinion present the greatest opportunities for sustainability improvements (e.g. based on possible innovation)?
- Which type of wind turbines (offshore or onshore) present the bigger challenge in term of sustainability?

Part 2: Reflection and Alignment with Lifecycle Phases

- Reflecting on the lifecycle phases, where do you see the greatest sustainability challenges?

- Which phase do you think is the most resource-intensive or costly in terms of financial trade-offs?
- Which KPIs do you consider the most critical and why (both financial and sustainability ones)?
- How do stakeholders (e.g., regulators, investors, operators) influence the focus on sustainability versus financial KPIs?
- Are there any lifecycle phases where you believe sustainability practices can be enhanced without significantly increasing costs?

Part 3: Literature Comparison and Confrontation

Sample Questions:

- The literature often emphasizes sustainability KPIs over the financial ones (e.g., EBITDA, OPEX). Does this align with your experience in the field?
- How would you describe the alignment between current policies and the industry's sustainability goals?
- Looking ahead, what trends or changes do you foresee in balancing sustainability and financial objectives in the wind energy sector?

A.2 Transcript

In the following section the transcript of the conducted interview will be presented. The part of the interviewer's speech (in this case myself) will be denoted in blue, while on the other hand, the part of the interviewee's responses will be denoted in red. The interviewee is a relevant industry expert working in one of the largest companies in the wind industry, specifically in the sustainability sector.

Interviewer:

Okay, so we're starting with the interview. First I want to ask you again, if you are fine with me recording this interview, transcribing it and use it for analysis in my bachelor thesis.

Industry Expert:

Yes, I am fine.

Interviewer:

Okay, thank you. Just once more to emphasize, the idea of the interview is to be like an open discussion.

So in this first part, we'll be discussing the sustainability and the trade-offs. So for the sake of the interview, could you again briefly describe what your expertise is and your role within the wind energy sector?

Industry Expert:

Sure thing. So my name is Andrew. I'm based in our group sustainability team here in Copenhagen, Denmark.

My background is in ecology, environmental sociology and I'm overall responsible for our commercial sustainability. It's basically how we bring more sustainable products to the market and our sustainability strategy as a group entity.

Interviewer:

Okay, thank you very much.

On to the next question. So in your view, what are the most pressing sustainability challenges in the wind energy sector today?

Industry Expert:

I think the biggest one in terms of scope would be the scope 3 reduction or the supply chain emissions reduction. And for us specifically, that looks like decarbonizing some of the material value chains that go into wind. So especially steel and iron materials, but also things like copper and aluminium.

And then the second biggest on the emission side would be the marine gas fuels. So the fuels we use for long distance transportation or for transportation of our vessels to and from offshore wind farms.

I think from a financial perspective, the biggest issue we faced is probably related to two

things. So either not having the right recycling solutions in place for composite materials, so think of like blades, or potential negative impacts on avian fauna, some birds and bats.

Interviewer:

Okay, thanks. Then are there specific life cycle stages where the challenges would be most notable? So which life cycle stage has the most obvious obstacles?

Industry Expert:

I think the biggest in terms of impact is going to be upstream. So the raw materials, mining and refining.

Interviewer:

So how do you perceive the interaction between sustainability and financial objectives in the wind energy sector? What would you say, are they aligned, interconnected or opposite to each other?

Industry Expert:

That is interesting topic. There is some alignment. So for instance, by reducing emissions, we can reduce exposure to ETS, carbon emission trading scheme taxes, as well as the carbon border adjustment mechanism, CBAM tariffs.

We also see projects that have non-price criteria in them. So for instance, when we're bidding on offshore wind auctions, especially in Europe, they often have part of the weighting of who is awarded that project is based on non-price criteria. So it could be things like biodiversity protection, it could be things like circularity, it could be things like reducing emissions. But our ability to have differentiated sustainable offerings impacts our competitiveness and winning those auctions.

And then of course, there are some things that are more difficult.

So if we talk about like we have a target to be carbon neutral by 2030 without using carbon offsets for scope one and two emissions. So that basically means that all of our vehicles, vessels, factories, offices, warehouses, et cetera, need to be running 100% renewable electricity or sustainable biofuels, which then in some cases is very difficult. So we've talked about like it's quite expensive to decarbonise offshore vessels, as an example. It can be difficult to switch completely to renewable vehicle or to electric vehicles in some markets, especially a lot of them in Asia Pacific for us. And then you have a trade-off on the cost, if you will, which isn't always necessarily justified by a premium or competitiveness.

Interviewer:

Then, so on to the next part. So which phase do you think is the most resource intensive or costly in terms of financial trade-offs?

Industry Expert:

In terms of financial trade-offs for Vestas specifically, I think it's going to be in the

construction and service phases. And that's where we sort of have the majority of our scope one and two emissions. And that's where we would face the majority of the cost in reducing those emissions.

If we talk about the wind industry, then I would still say the same as before, it's going to be the raw material mining and refining stages.

Interviewer:

In terms of sustainability improvement, in your opinion, do you see in future or even nowadays some new innovations or practises which could make a significant difference and have significant impact?

Industry Expert:

Yes. So on the emissions side, we actually have an offering now on what's low-emission steel, which is basically a recycling of scrap-based steel using on-site renewable electricity to power an electric arc furnace. So by doing that, we can reduce emissions from the wind turbine by about 50% compared to using conventional steel.

A little bit in the future, you also have work on what's called near-zero emission steel, which would almost completely decarbonise steel production. And that's based on using hydrogen as a reduction agent in the steelmaking process instead of coal. So it opens up some new possibilities. Because it doesn't depend on scrap, you can make new steel that way, and it also further reduces the emissions beyond scrap steel.

On the circularity side, we also, of course, have an ongoing project with SSStena Recycling and Ölin Epoxy based around building up a circular economy marketplace for composite materials. So that would be things like fiberglass, carbon fiber, metals, resins, etc., which is a new thing in the world. So composites aren't typically recycled today because they're difficult to recycle. So we've invented the technology to enable that recycling, patented it, and now we're working on building that marketplace for those kinds of recycled materials.

Interviewer:

Then, which KPIs do you consider the most critical and why? Maybe we can start with the sustainability one and then cross on to the financial ones. So the ones which you track usually.

Industry Expert:

Yeah, I think they're the straightforward ones in the industry. So scope one, two, three emissions. And then for circularity, we have one called material efficiency, which is basically defined as the tonnes of waste excluding recycled.

So basically we're trying to reduce that material efficiency ratio over time by either making sure that the waste that we do produce is in recyclable form and it's actually recycled, or that we just reduce the amount of waste overall. This would probably be the two main ones for us as a manufacturing company, or it's part of the two main categories. So the three scopes of emissions and then the material efficiency.

Interviewer:

Okay, and the financial ones, do you have some specific which you track also?

Industry Expert:

I think the two main financial ones would be the reducing our exposure to carbon related taxes and tariffs. So usually on the ETS receipt and exposure and then improving project margin. For instance, when we sell low emissions deal, if we sell that as a margin, then that can also be one way to sort of improve the emissions, sorry, improve the project financials by sort of selling sustainability at a premium cost.

Interviewer:

Okay, so the next question would be how do the stakeholders, so let's say regulators, investors, influence the focus on sustainability versus the financial KPIs? Do they have a saying?

Industry Expert:

Yeah, of course. So I think they drive a lot of it actually. Primarily you can say that they have a market driven mechanism, which is the emission trading scheme and carbon border adjustment mechanism, and then they have usually a reward scheme, which is sort of non-price criteria and auctions.

So the combination of those two is as they try to influence the market to be more sustainable.

Interviewer:

Are there any life cycle stages where you believe that sustainability practises can be enhanced without significantly increasing any costs?

Industry Expert:

No. I mean, I think those low hanging fruits have, you know, mostly been taken. So the stuff that hasn't been done is mostly related to increased costs now.

Interviewer:

I would like to talk a bit about the findings I found in the literature and how they reflect the real situation.

So in the literature which I read, they often, when they talk about sustainability and KPI trade-offs, they often emphasise the sustainability side. So the EBITDA or the OPEX or such measurements are underrepresented in comparison to the sustainability ones. So does this also align with your experience in the field?

Is the emphasis mostly on the financial side or on the sustainability side when we talk about the trade-off, in your opinion?

Industry Expert:

I think the financial side is a prerequisite, right? So if you don't have a financially

sustainable project, it's just not going to happen. So of course, that's where the waiting is.

Yeah, I think it's pretty clear in any business, if the financials don't work, then you don't have a company. So that's the priority.+

But I think maybe the challenge, if you will, of trying to be a sustainable company or being a sustainable company is defining the thresholds at which you won't engage in a project based on non-price issues. So it could be, for instance, human rights issues on a project, or it could be that's too close to a really biodiversity-sensitive area. So sort of defining those exclusion areas where we sort of don't operate no matter what is maybe the more interesting trade-off.

And you can see that as an ESG (Environmental, social, and governance) and financial trade-off, or you can see that as a risk and financial trade-off, right? Because having projects in those kinds of areas or with those kinds of risk profiles also has a financial impact. So I don't necessarily see them as completely separate, right?

So yeah, at least in some cases, the risk picture, in other words, the threat that a project won't be developed or will be paused from development or will have legal challenges, it's related to the ESG profile.

Interviewer:

In your opinion, are the sustainability obstacles more obvious within the lifecycle of offshore or onshore turbines?

Industry Expert:

Offshore, I think just because the decarbonizing offshore logistics is so difficult, because it still relies on marine gas oil permanently.

Whereas with onshore, you have a lot of innovation around electric vehicles and trucks and batteries, which will open up more technological possibilities than we see offshore.

Interviewer:

Yeah, understandable. Then the next question would be, so how would you describe the alignment between current policies and the industry's sustainability goals? Meaning, are the current sustainability goals, so when we talk about the policies, let's talk maybe about, so we can have in mind the Paris Climate Agreement or such agreements.

So are they aligned? Is it even possible to catch those goals?

Industry Expert:

Yes, yeah. The policies are not in place to meet the goals of the climate agreement.

Interviewer:

Okay. And lastly, so looking ahead, what trends or changes do you foresee in balancing sustainability and financial objectives in the wind energy sector in the future?

Industry Expert:

I think right now, the big push in the last few years has just been a lot more transparency and seriousness around the ESG figures. So you see a lot of legislation has been focused on sort of forcing more reporting and more auditing of ESG figures, trying to make those numbers better. So I think that now that some of that's in place, companies are sort of forced to take the next question, which is like, what do we do with that information? So now we have better information. Do we want to use that as a strategic lever? And I think some companies see having a strong sustainability profile as sort of increasing their ability to market certain types of consumer groups or to access certain markets, so for us, like non-price criteria and auctions, or just de-risking their future operations. So if they foresee that energy costs are going to rise a lot, or material costs are going to rise, or carbon-related taxes, tariffs are going to rise, you can say that. Focussing on those areas is just part of competitiveness as on a cost basis. I don't know if that answers your question, but it's a- Yeah, yeah, mostly.

Interviewer:

So yeah, those would be all the questions covered. Is there something you want to add in terms of maybe I didn't cover something with my questions, maybe something else is interesting by chance?

Industry Expert:

Maybe something for wind specifically, which is interesting, is this question of because wind is already sort of the most sustainable form of electricity, we're often asked why would you increase the price of the cost of wind to make it even more sustainable when you should say maybe that doing it at any cost and as cheaply as possible to try to force out fossil fuels out of the market would be the more sustainable approach, right? So I think that's an interesting trade-off, which we haven't really considered, but you can say that sort of doing everything you can to make wind energy as cheap as possible be the most sustainable thing to do because that would help it penetrate the market further, right?

So anything that increases cost on wind could be seen as actually an impediment to sustainability. So an interesting perspective to take, not one that we take personally, but yeah

Interviewer:

Okay, thank you very much again for participating in this interview. I'm going to stop the recording now.